DESIGN THINKING

IIITA Registration Experience Empathize Stage Report

First year BTech admission registration journey

Stage	Empathize
Research population	First year BTech students at IIITA
Response period	3 September to 8 September 2026
Prepared for	Design Thinking project evaluation

Core finding

Students usually completed registration, but success often depended on peer guidance and improvised workarounds. The strongest opportunity is a single, reliable journey that confirms requirements before travel, supports uploads before arrival, guides students step by step on campus, and hands over a clean, ready hostel room.

Executive Summary

This report analyzes open text responses from first year BTech students about financial registration, academic verification, hostel allocation, mess setup, and related admission formalities at IIITA. The purpose is to understand what students did, thought, felt, and needed during the journey rather than to judge individual staff members or systems.

The dataset contains 33 submitted rows. One exact duplicate and two blank rows were removed, leaving 30 unique records. Completion varied by question, so coded frequencies are directional signals, not survey prevalence estimates. The analysis emphasizes recurring experiences, workarounds, emotional effects, and unmet needs.

Main Findings

- Unclear or late document requirements created the highest stakes anxiety because students had already started travelling or had limited return travel options.
- ERP problems included crashes, repeated logins, unclear upload limits, delayed fee confirmation, and difficulty uploading documents on campus when the network was weak.
- Senior and peer WhatsApp groups often acted as the practical help desk because they were accessible, responsive, and able to explain the next step in familiar language.
- Queues, token uncertainty, insufficient seating, and unclear movement between desks made an already emotional arrival feel tiring and chaotic.
- Dirty rooms, broken fixtures, heat, water concerns, and last minute maintenance delayed the transition from applicant to settled resident.

Design Direction

The evidence supports a joined registration service rather than isolated fixes. Students need one verified checklist, an upload readiness check, a visible on campus route and queue status, trained guides, preassigned hostel rooms, and a simple way to report room defects. The service should preserve the useful human support now provided informally by seniors while moving authoritative information into an official channel.

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Research Scope and Process

Research Question

How do first year BTech students experience IIITA registration from preparation at home through academic and financial verification to settling into the hostel and mess, and where do they need clearer information, faster support, or better physical readiness?

Participants and Data

Item	Recorded value	Treatment in analysis
Submitted rows	33	Starting dataset
Exact duplicate	1	Removed
Blank rows	2	Removed
Unique retained records	30	Included where an answer was interpretable
Collection period	3 to 8 September 2026	Used as fieldwork window
Personal identifiers	Names and roll numbers supplied	Excluded from this report

Empathize Activities

1. Prepare open ended questions in related themes so students can describe events in their own words.
2. Collect accounts of preparation, portal use, physical registration, hostel entry, emotions, workarounds, and desired solutions.
3. Clean the response sheet by removing duplicate and blank submissions while retaining partial responses as qualitative evidence.
4. Code statements into recurring themes and separate reported events from inferred needs.
5. Map the journey, build an empathy map, express Jobs to Be Done, and form composite personas from recurring patterns.
6. Translate the findings into opportunity areas for the Define and Ideate stages.

Ethical Handling

The report anonymizes all respondents. Short evidence excerpts are used without names or roll numbers. The personas are composites built from several responses and must not be interpreted as profiles of named students. The team should retain the original response sheet securely and avoid sharing it outside the project group.

Our Contribution

We completed the data cleaning, initial theme clustering, wording refinement, and document production ourselves. We did not alter the interview responses. Before submission, we compared the coding with the

source sheet, agreed on disputed interpretations, and added field notes or observations that were not captured in the form.

Interview Questionnaire

Use the following guide for follow up interviews. Ask the main question first, listen without interrupting, and use the probes only when they help the participant describe a concrete event. Avoid suggesting an answer. With consent, record the interview or take detailed notes.

Arrival and Expectations

1. Tell me about the moment you arrived on campus for registration. What were you expecting to happen that day?
2. What were you most excited, worried, or uncertain about before the process began?
3. Why did those expectations matter to you or your family?

Preparation Before Travel

4. Walk me through how you learned which documents, payments, and forms to prepare before travelling.
5. Which requirement was hardest to understand or arrange, and what made it difficult?
6. Tell me about any information that changed or arrived late. What did you do next?
7. Where did you check whether you were fully ready, and why did you trust that source?

Financial and Academic Registration

8. Show or describe the steps you followed for fee payment, document upload, and academic verification.
9. At which step did you slow down, repeat work, or ask for help? What caused that?
10. How did you know a payment or uploaded document had been accepted?
11. What happened when the portal, network, or equipment did not work as expected?
12. Why do you think that step became difficult for you?

On Campus Movement and Support

13. Describe where you went from the moment you entered the registration area until you received your enrollment or roll number.
14. At each handoff, how did you know where to go and how long you might wait?
15. Tell me about a moment when a staff member, senior, parent, guard, or peer helped you move forward.
16. What did you do when different people or channels gave different answers?

Hostel and Mess Transition

17. Walk me through room allocation, key handover, room entry, and mess setup.
18. What condition was the room in when you first entered, and what did you have to arrange yourself?
19. Which issue most affected your ability to rest, study, eat, or feel settled?
20. Why was that issue important during your first two days?

Workarounds and Nonconsumption

21. What unofficial group, person, website, or workaround did you use to complete a task?
22. Why did that option work better than the official instruction or service?
23. Was there any official resource you stopped using or never used? Tell me why.
24. Which task would you prefer to avoid entirely, and what outcome would still need to be guaranteed?

Desired Experience

25. Imagine the process worked exactly as you needed. What would happen before arrival, on campus, and at the hostel?
26. What information or confirmation would make you feel confident at each stage?
27. If only one part could change next year, which part would you choose and why?

Interview Observation Sheet

Complete this sheet during or immediately after each interview. Record what happened rather than only whether the participant liked or disliked the process.

Observation area	What to capture
Context	Travel situation, companions, time pressure, device and network access
Actions	Sequence followed, repeated steps, documents handled, people approached
Signals	Pauses, visible confusion, confidence, fatigue, relief, disagreement
Breakdowns	Exact point of failure, information available, consequence, recovery
Workarounds	Unofficial channel or improvised method and why it was chosen
Evidence	Participant wording, screenshot description, form or portal step
Interpretation	Possible need or insight, clearly marked as the interviewer’s interpretation

Recommended Follow Up Sample

Select six to eight students from contrasting experiences: at least two who reported a smooth process, two who faced ERP or network trouble, two who encountered document uncertainty, and two who experienced hostel readiness problems. Include students who travelled from far away or faced fixed return travel because time pressure changes the impact of uncertainty.

Interview Conduct Checklist

- Ask permission before recording or photographing any evidence.
- Use open questions and request a specific example when an answer is general.
- Ask why with care. Use wording such as What made that difficult or Why did that matter to you.
- Keep questions from the same theme together before moving to the next stage of the journey.
- End by summarizing what you heard and asking the participant to correct the summary.

Response Profile and Evidence

The responses ranged from detailed narratives to single word answers. Some students described a largely smooth journey, while others reported several connected breakdowns. The table below shows directional coded frequencies across the 30 retained records. A record is counted when it contains an interpretable statement related to the theme. Because some fields were blank, the values should guide prioritization rather than support population estimates.

Theme	Directional count	Typical evidence
ERP portal or network friction	about 15	Crashes, repeated login, upload errors, unclear size limit, delayed fee status, weak on campus network
Document clarity or verification burden	about 14	Late or missing requirements, notarization confusion, repeated checking, too many documents
Reliance on seniors or peers	at least 16	WhatsApp groups, batchmates, seniors, friends, or other students used for guidance
Hostel room or basic facility readiness	about 12	Dirty room, broken lock or fittings, painting, heat, water, mess, sharing mismatch
Queue crowding or token uncertainty	about 9	Insufficient seating, slow desks, crowding, lack of line, tokens ending
Wayfinding or staff guidance gap	about 8	Unclear next desk, insufficient guides, conflicting or incomplete explanations

Evidence Excerpts

Theme	Anonymized excerpt	Interpretation
Document clarity	One document was not mentioned in the advisory and it was asked here	The checklist was not experienced as complete or final
ERP reliability	The website crashed without warning and constantly needed a new login	Failure recovery added repeat work and uncertainty
Peer support	Seniors personally guide us	Students valued responsive, situation specific help
Queue management	There was no line and a lot of chaos	Students lacked a predictable service order
Hostel readiness	The room was quite dirty and there was a problem with the fan light and door lock	Room condition delayed settling and transferred setup work to families

Positive Signals

Several students reported easy fee payment, smooth room allocation, supportive officials, or quick settling. These successful conditions should be preserved.

Registration Journey

Stage	Student action	Breakdown	Underlying need
Prepare at home	Read advisory, arrange documents, pay fees, plan travel	Late or unclear documents; uncertainty about credentials and formats	Confidence that the checklist is complete
Upload and confirm	Log in, upload documents, check payment and verification status	Crashes, re login, file size ambiguity, delayed status, weak network	Clear acceptance status and recovery guidance
Arrive and orient	Find venue, wait for token, attend briefing, locate next desk	Crowding, heat, limited seating, unclear sequence, tokens finishing	Visible route, expected wait, and accountable help
Verify and enroll	Present documents, correct errors, obtain ID or roll number	Repeated checks, inconsistent answers, slow upload desks	One verified record and fewer repeated handoffs
Enter hostel and mess	Receive room, inspect facilities, clean, arrange food and water	Dirty or unfinished rooms, broken fixtures, heat, water and mess concerns	A safe, clean, functional room on arrival
Settle into campus	Say goodbye to family, meet peers, begin activities	Fatigue and unresolved room issues can prolong unease	Early belonging, rest, and confidence about where to seek help

Emotional Pattern

Students often began with excitement, pride, nervousness, or sadness about leaving family. Confidence dropped when a requirement changed, the portal failed, or the service order was unclear. Relief returned when a peer or official provided a definite answer and when the student received a room. Room defects, heat, or poor water could then create a second dip after registration appeared complete.

Critical Moments

- The final checklist before travel determines whether students and families feel prepared.
- A portal confirmation must distinguish uploaded, accepted, rejected, and action required states.
- The first on campus handoff should explain the complete route, current wait, and exception process.
- Hostel key handover is not the end of the journey. The room must be ready for immediate use.

Empathy Map

This map represents the shared experience visible across the retained responses. It combines recurring evidence and does not describe every student.

Says ● I expected the process to be smooth and fast ● Everything should be mentioned beforehand ● Seniors were more accessible and answered every question ● The room was dirty or something was broken	Thinks ● Have I brought every document in the correct format ● Will the portal keep my upload and payment status ● Where do I go next and will my token be accepted today ● Can my family leave knowing I am settled safely
Does ● Checks advisories, email, ERP, WhatsApp, YouTube, Reddit, and peer messages ● Reuploads files, restarts sessions, asks several people, or uses personal internet ● Waits in queues and moves between desks with documents ● Cleans the room, buys a cooler, or reports broken fixtures	Feels ● Excited and proud about joining IIITA ● Nervous about leaving home and making a mistake ● Frustrated or tired when information or systems fail ● Relieved when a trusted person confirms the next step

Needs

- A definitive and current list of documents, formats, examples, deadlines, and notarization rules
- A portal that explains errors and preserves progress under heavy traffic or weak connectivity
- An official source that is as quick and approachable as peer messaging
- A predictable on campus sequence with visible queues, seating, ventilation, and exception support
- A clean functional room with accurate occupancy information and rapid defect resolution

Central Insight

Students did not only need tasks to be completed. They needed repeated proof that they were on the right path. When official systems did not provide that proof, they borrowed confidence from seniors, peers, guards, and family members.

Pain Points and Needs

Pain point	Observed signal	Effect	Need	Priority
Late or ambiguous requirements	Unexpected affidavit, undertaking, notarization, or unnamed document	Emergency preparation, fear of rejection, family and travel disruption	A version controlled checklist with examples and change alerts	Very high
Unreliable digital completion	ERP crash, repeat login, upload ambiguity, delayed payment status	Repeated work and dependence on local connectivity	Resilient uploads, clear errors, saved progress, receipt and status	
Fragmented assistance	Students ask seniors or several officials for the same answer	Conflicting guidance and uneven access to help	One official help channel with trained human escalation	
Crowding and uncertain queues	No clear line, limited seating, token limits, heat	Fatigue, frustration, risk of an extra hotel night	Arrival slots, live token status, seating, signage, overflow plan	
Repeated verification	Online upload followed by several physical checks	Time cost without a clear reason for duplication	One verified record and explicit exceptions requiring originals	
Room not ready	Dirt, painting, broken fan light lock, heat, water or mess concern	Student cannot rest or feel settled after a tiring day	Pre handover inspection and rapid maintenance ticket	

Root Causes Suggested by the Evidence

- Information appears across multiple channels without a clearly experienced source of truth.
- Digital steps are treated as complete even when students cannot see a durable confirmation state.
- The physical process exposes internal handoffs to students without enough wayfinding or queue visibility.
- Hostel readiness is separated from registration completion even though students experience both as one arrival journey.
- Peer networks fill a service gap but do not have formal access to verified current information.

Problem Statement

First year students and their families need a registration journey that confirms preparation before travel, explains every next step during campus reporting, and provides a ready place to stay because uncertainty, repeated verification, and room defects make an important transition unnecessarily tiring and dependent on unofficial help.

Jobs to Be Done

Jobs to Be Done describe the progress students seek in a situation. They do not prescribe a particular portal, desk, or app. Each job below combines functional progress with the emotional reassurance visible in the interviews.

Job	Job statement
Prepare with confidence	When admission approaches, help me know the exact current requirements and acceptable formats so I can leave home certain that no missing document will delay enrollment.
Complete digital work before travel	When I have access to my documents and a stable connection, help me upload, correct, and confirm them before arrival so campus time is used only for steps that require physical presence.
Recover from an exception	When a payment, upload, or document is rejected, help me understand the reason and the next valid action so I do not repeat the whole process or ask several people.
Navigate the campus process	When I arrive with my family, help me see where to go, what to carry, and how long each step may take so I can plan the day and avoid unnecessary waiting.
Receive trusted human guidance	When instructions are unclear or changing, connect me with a responsive person who has verified information so I can act without relying on rumor.
Move into a ready room	When registration is complete, give me an assigned, clean, functional room and a fast defect reporting path so I can rest, study, and let my family leave with confidence.

Functional Emotional and Social Dimensions

Dimension	Desired progress	Evidence in responses
Functional	Finish required tasks correctly with minimal repeat work	Requests for a complete checklist, reliable ERP, faster verification, and room readiness
Emotional	Feel certain, supported, and able to settle after leaving home	Nervousness, family separation, fatigue, frustration, relief after guidance
Social	Appear prepared to family and enter the student community confidently	Parents travelling with students, reliance on batchmates and seniors, desire to meet peers

Nonconsumption and Workarounds

- Some students did not use official guides because WhatsApp groups answered faster or translated instructions into concrete steps.
- Students used personal hotspots or ethernet when campus connectivity did not support portal work.
- Students asked classmates after instructors could not resolve an upload or document issue.
- Students cleaned rooms, bought coolers, or waited for repairs because readiness checks had not completed the job before handover.
- Some students wanted to skip on site uploading, repeated document verification, or a long orientation that did not provide the expected campus guidance.

Composite Personas

The following personas are design tools built from recurring response patterns. They are not demographic segments and do not represent named respondents.

The Prepared Planner

Attribute	Description
Situation	Travels with family and faces fixed return travel or accommodation plans
Behavior	Reads advisories, pays fees, uploads early, and plans the reporting day
Goal	Finish as planned and let family return without disruption
Pain	A late requirement can invalidate careful preparation
Need	A final checklist, change log, readiness score, and reporting slot
Representative thought	Tell me the complete requirement before I start travelling

The Guided Newcomer

Attribute	Description
Situation	Is new to campus and wants reassurance about each next step
Behavior	Asks seniors, peers, guards, or officials and follows WhatsApp guidance
Goal	Know what to do without feeling lost or making a mistake
Pain	Official guidance can feel slow or inconsistent

Attribute	Description
Need	Trained guides, official chat, visible signs, and completion cues
Representative thought	I trust the person who answers my exact question now

Composite Personas Continued

The Digitally Blocked Registrant

Attribute	Description
Situation	Has the required documents but faces a portal, device, file, payment status, or network failure
Behavior	Retries, logs in again, changes the file, uses a hotspot, or asks peers for a workaround
Goal	Complete the task without losing progress or being blamed for a technical failure
Pain	Errors do not explain the cause or preserve a recoverable state
Need	Specific error messages, automatic save, upload validation, traffic resilience, and a support ticket with evidence
Representative thought	Show me what failed and let me continue from that point

The Transitioning Resident

Attribute	Description
Situation	Has completed enrollment and is emotionally moving away from home for the first time
Behavior	Inspects the room, cleans, arranges a cooler or supplies, seeks food and water, and contacts the warden
Goal	Rest, feel safe, begin making friends, and reassure family that basic needs are met
Pain	Dirt, unfinished maintenance, broken fittings, heat, water, food, or occupancy surprises
Need	Accurate room assignment, readiness inspection, essentials information, and a rapid maintenance response
Representative thought	Registration is not finished until I can actually settle into the room

Persona Coverage

A student may move through several personas in one day. The personas should guide service states and recovery paths rather than assign permanent labels.

Opportunity Areas

Opportunity	Service concept	Measure in the next cycle
One verified readiness checklist	Publish a dated checklist with plain language names, samples, format and size rules, notarization status, owner, and change history	Checklist comprehension test and fewer day of admission document exceptions
Prearrival registration workspace	Open credentials several days early, validate files before upload, save progress, and issue a downloadable readiness receipt	Upload completion, error recovery, and confirmed readiness before travel
Guided campus route	Use assigned arrival slots, live token status, clear signs, a printed route, seating, ventilation, and an exception desk	Wait time, abandoned tokens, repeat visits, and wayfinding questions
Official help with peer reach	Create an official messaging channel supported by trained staff and designated student volunteers using the same verified knowledge base	First response time, resolution rate, repeated questions, and conflicting answers
Single verification record	Show which evidence is already verified and request originals only where policy requires them	Number of checks and average handling time per student
Ready room handover	Preassign rooms, inspect cleanliness and fixtures, disclose occupancy, and open a QR based maintenance ticket at the door	Rooms passing inspection, defects at handover, time to resolution, and first night relocations

Prioritization

1. Stabilize document requirements and prearrival confirmation because a late requirement can prevent completion and disrupt travel.
2. Improve portal recovery and status visibility because technical uncertainty creates repeat work at scale.
3. Create a clear on campus route and accountable help point because these changes can reduce confusion even before a full system redesign.
4. Treat room readiness as part of registration completion because the first night strongly shapes the sense of arrival.

How Might We Questions

- How might we prove to a student before travel that every required document is current and acceptable?
- How might we let students recover from an upload or payment exception without repeating completed work?
- How might we make the next step and expected wait visible throughout on campus reporting?
- How might we combine the trust and accessibility of senior groups with verified official information?
- How might we make hostel room readiness a visible condition of completing registration?

Limitations and Validation

Study Limitations

- The sample was gathered through a form and may overrepresent students who had access to the distribution channel.
- Question completion varied, and some answers were too brief to code confidently.
- The form captures recalled accounts rather than direct observation of the registration environment.
- Directional theme counts depend on interpretation and were not independently coded by two researchers.
- The responses do not include parent, staff, warden, mess contractor, or admissions administrator perspectives.
- The study describes the 2026 experience and should not assume that every reported issue persists unchanged.

Validation Plan Before Define Stage

1. Have two group members independently code the 30 retained records and reconcile differences.
2. Conduct six to eight follow up interviews using the refined guide and collect concrete examples of failures and successful recoveries.
3. Observe one registration simulation or reconstruct the route with staff to record handoffs, documents, queue points, and exception owners.
4. Interview admissions, ERP, hostel, and mess stakeholders to test constraints behind repeated checks and late information.
5. Create a simple prototype of the readiness checklist and journey tracker, then test whether new students can complete realistic tasks without help.

Suggested Evidence Pack

- An anonymized response sheet with a codebook and a record of excluded rows
- Interview notes and consent record for follow up participants
- Photographs or sketches of the physical route without identifiable faces unless consented
- Screenshots of portal states with personal data removed
- A revised journey map, empathy map, Jobs to Be Done, and persona set after team validation

Completion Criteria for Empathize Stage

The stage is ready for review when the group can trace each major insight to evidence, explain differences among student experiences, show how workarounds reveal unmet needs, and state which assumptions still require validation. A polished output should preserve the uncertainty in the data rather than present every theme count as a precise population statistic.

Appendix Coding Framework

Use these codes consistently when reviewing the original response sheet or adding follow up interviews. A response may receive more than one code.

Code	Theme	Include when the response mentions
DOC	Document clarity	Missing, late, ambiguous, notarized, repeated, or excessive documents
ERP	Portal reliability	Crash, login loop, upload failure, unclear limit, status delay, confusing navigation
NET	Connectivity	Weak campus network, personal hotspot, ethernet, unusable computers
QUE	Queue and capacity	Crowding, token shortage, slow desk, limited seats, heat in waiting area
NAV	Wayfinding	Unclear next step, location, sequence, or responsible person
PEER	Peer workaround	Senior group, batchmate, friend, social media, YouTube, Reddit
STAFF	Official support	Helpful official, insufficient staff, conflicting answer, uninformed instructor
HOST	Hostel readiness	Cleanliness, broken fixture, painting, occupancy, room allocation, heat
MESS	Mess and water	Food, drinking water, bathing water, mess setup
EMO	Emotional effect	Excitement, pride, anxiety, sadness, fatigue, frustration, relief, belonging
TIME	Time and travel consequence	Return train, hotel night, reporting slot, delay, family schedule
POS	Positive condition	Smooth task, supportive person, successful preparation, quick settling

Evidence Standard

For each code, store the original statement, the interviewer’s short interpretation, and the participant context that changes the impact. Do not infer intent from a single word when the meaning is unclear. Mark uncertain codes for team review. When counting themes, count each retained respondent once per theme even if the theme appears in several answers.

Recommended Team Roles

Role	Responsibility
Interviewer	Ask open questions, probe concrete events, and obtain consent
Observer	Record behavior, environment, sequence, and nonverbal signals
Coder	Apply the agreed codebook without adding unsupported meaning
Reviewer	Check anonymization, evidence traceability, and disagreement resolution
Synthesizer	Update maps, jobs, personas, and opportunity statements after validation

Final Review Checklist

- At least 15 students are included and the retained dataset is documented.
- Questions are open ended, organized by theme, and include thoughtful why probes.
- Pain points are linked to events, consequences, needs, and supporting evidence.
- Empathy map, Jobs to Be Done, and personas are consistent with the source data.
- Personas are labeled as composites and personal identifiers are removed.
- We completed all work in this project and checked all interpretations as a team.
- The next stage begins with prioritized How Might We questions rather than a fixed solution.